

DICKSON TAWIAH AMAN , ARAeS

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Portfolio: Travan01.github.io/dickson-portfolio/

PROFILE

Multidisciplinary Engineer with proven expertise in Aerospace and UAV design, Mechanical & Electrical Systems, Embedded Systems, Cybersecurity, and Sustainable Engineering. Passionate about creating innovative, eco-efficient solutions that boost mechanical and electrical performance. Accomplished in engineering design, system analysis, and cross-functional collaboration, consistently delivering projects on time and to specification. Seeking roles in Aerospace, Mechanical, Electrical, Systems, or Maintenance Engineering.

CORE SKILLS

Technical

- Mechanical Design & Simulation: Siemens NX, Solid Edge, SolidWorks, ANSYS, AutoCAD, CATIA, Revit, Simulink
- Electrical & Embedded Systems: CNC programming (G/M codes), Python, Java, Kotlin, MATLAB, C/C++, Arduino, STM32, PLC, Linux, Proteus, KiCad, PCB design
- Sustainable Engineering: Renewable-energy systems, Net Zero systems design, sustainable irrigation solutions
- Data & Analytics: Microsoft Excel, Power BI
- Hands-on: Metalwork, carpentry, painting, machine assembly, engine servicing, CNC setup

Professional

Leadership • Team Collaboration • Project Management • Technical Training • Problem-Solving • Clear Communication

EXPERIENCE

Senior Shift Engineer — The Stratford (Oct 2025– Present)

- Lead engineering operations across hotel mechanical, electrical, HVAC, and plumbing systems.
- Perform advanced diagnostics, corrective maintenance, and system optimisation on plant equipment.
- Ensure compliance, safety, and asset reliability while supervising contractors and junior engineers.
- Manage shift-based engineering logs, BMS monitoring, and emergency response procedures.

Maintenance Engineer — Whitbread Plc (Dec 2024 – Nov 2025)

- Complete preventive and corrective maintenance across electrical, HVAC, plumbing, and automated systems.
- Troubleshoot PLC-controlled plant-room faults and complex system failures.
- Use CMMS to manage tasks, compliance, and reporting.
- Supervise external contractors and manage maintenance budgets.

Freelance Technical Consultant — CTN Technologies (Mar 2023 – Oct 2024)

- Delivered UAV design, maintenance, and pilot training to civil and defence clients.
- Provided drone adoption strategies for improved efficiency and sustainability.
- Authored SOPs and training modules for UAV maintenance and flight data analysis.

Support Engineer — Ollennu Farms & Investments (Sep 2021 – Oct 2023)

- Diagnosed and repaired PLC-controlled processing-line faults.
- Operated agricultural drones for surveying and precision fertiliser application.
- Re-engineered tools, pumps, motors, and conveyor systems to improve reliability and throughput.

Aircraft Maintenance Apprentice — Ghana Air Force Aviation College (Nov 2021 – Nov 2023)

- Gained hands-on experience across aircraft mechanical, avionics, hydraulic, fuel, and pneumatic systems.
- Conducted inspections, preventive maintenance, and safety checks to military standards.

EDUCATION

BSc Aerospace Engineering — First Class Honours

Kwame Nkrumah University of Science and Technology, Ghana (Jan 2021 – Aug 2024)

UK Framework: RQF Level 6 / SCQF Level 10 / CQFW Level 6

Key Modules: Aerodynamics, Propulsion, Thermodynamics, Fluid Mechanics, CAD & Manufacturing, CNC Programming, Renewable Energy, Control Systems, Aerospace Vehicle Design, Programming for Engineers.

Affiliation: Associate (ARAEs), Royal Aeronautical Society

Awards: Certificate of Honor – UAV Training • Best Academic Tutor 2023/24 • Best Final Year Project – Hybrid Engine Development

PROJECTS

- Hybrid Turbofan Engine Control & Sensor System – Designed a sustainable hybrid propulsion concept with automated fuel-management and sensor integration.

- Radar Surveillance System – Built an ultrasonic radar with GUI visualisation and optimised energy performance.
- RC VTOL with Obstacle Avoidance – Developed a tri-copter VTOL with path avoidance algorithms.
- Rocket Flight Computer – Engineered a flight computer with altitude tracking and safe-deployment logic.

CERTIFICATIONS

- Professional Software Programmer (C & C++)
- UAV Training Certification
- Principles of Occupational Health & Safety Management
- Cybersecurity: Threats & Attack Prevention
- Ghana Air Force Practical Maintenance Engineering Training
- FMVA® Certification
- UK CAA Flyer ID (A1&A3) (2025–2030)
- UK CAA Operator ID (2025–2026)
- UK ENIC Statement of Comparability Certificate